Self-Supervised Classification of SAR Images With Optical Image Assistance

Chenxuan Li, Weiwei Guo, *Member, IEEE*, Zenghui Zhang, *Senior Member, IEEE*, and Tao Zhang, *Member, IEEE*

***Abstract*— Supervised deep neural networks (DNNs) have proven to be powerful tools for synthetic aperture radar (SAR) image interpretation tasks. However, they present a formidable challenge in acquiring a substantial amount of labeled data. In this article, we investigate the promising technique of contrastive self-supervised learning for SAR image classification. This approach allows us to take advantage of a large number of available unlabeled images to pretrain an SAR image classification model. Our novel contrastive learning framework conducts both instance-level and cluster-level pretext tasks, which not only enforce consistency between the images and their augmented "views" at the instance level but also their representation within clusters. Besides generating different views through random, low-level image transformations, we proposed two new strategies to construct positive sample pairs to improve contrastive SAR image feature learning: middle-level optical assistance and high-level graph searching. The middle-level optical assistance strategy is inspired by the observation that domain experts typically interpret SAR images with the aid of optical images. This insight spurs us to generate intermediate SAR images as positive samples from geographically matched optical data using CycleGAN. Furthermore, we augment the positive samples of the image with their k-nearest neighbor (KNN) counterparts, following the idea that the KNN samples should belong to the same cluster. Extensive experimental results conducted on the SEN12MS land cover classification benchmark dataset demonstrate that our method is competitive with state-of-the-art self-supervised methods for SAR image classification. Even with only a small amount of labeled data for fine-tuning the model, our method rapidly improves classification performance, surpassing models pretrained on natural image datasets.**

***Index Terms*— Contrastive clustering (CC), image fusion, land cover classification, self-supervised learning (SSL), synthetic aperture radar (SAR).**

I. Introduction

SYNTHETIC aperture radar (SAR) is a type of active microwave imaging radar that can continuously and dynamically observe the Earth under various weather and lighting conditions, having unique advantages over other sensors. Over the years, numerous SAR platforms have provided massive data for various earth observation tasks, including urban planning, land use and land cover monitoring, agriculture, and so on. However, due to geometric distortions and strong speckle noise, traditional handcrafted features for SAR image classification are weak and fragile. During the latest decade, deep neural networks (DNNs) have proven to excel at learning highly effective feature representation from raw data for image classification [1], which also achieved significant success in various SAR image interpretation tasks [2]. However, these performance improvements mainly stem from the usage of supervised learning methods that require extremely large-scale labeled data.

SAR images reflect the electromagnetic backscattering characteristics of the surface of objects on the ground. Due to the special imaging mechanism and the resulting geometric distortions and speckle noise in SAR images, the visual understanding of SAR images becomes much more complicated and difficult [3]. It is impractical to collect and annotate a large number of SAR images. Over the years, the self-supervised paradigm [4] has emerged to overcome the need for large labeled data. Self-supervised methods learn useful feature representation leveraging numerous unlabeled data, and sequentially, the models are fine-tuned for downstream tasks. Self-supervised learning (SSL) methods carefully design elaborate pretext tasks to train the models where the supervised signals are generated by utilizing data transformations and characteristics of the data itself, such as spatial context structure and content similarity [5]. Some recent work performed representation learning and cluster simultaneously, which applied cluster labels by k-means as the supervision signal to train the network [6]. To enforce feature representations and cluster performance, some methods [7], [8] introduced contrastive learning into clustering. The basic idea of contrastive learning is to learn representation by maximizing the similarities between different views or augmented versions of the same input while minimizing those of different inputs. The keys to the success of contrastive feature learning are to construct contrastive pairs and design the contrastive loss.

Self-supervised techniques have been adopted for remote sensing (RS) image classification [9], [10], semantic segmentation [11], and change detection [12], [13]. However, most of the existing works only deal with single-modality data, such as optical or SAR data. Little attention was paid to applying SSL techniques to process multimodal RS image data. Due

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to the special mechanism of microwave active imaging, visual cognition of SAR images is very difficult, so that the domain experts usually interpret the SAR images aided by the optical images with some physical prior knowledge of the scattering and imaging mechanism of the objects. It indicates that optical images can provide potential complementary cues to SAR image understanding and are beneficial to extracting more subtle SAR image features for classification.

To process multimodal data, recent studies consider the fusion of SAR and optical images, thus improving classification accuracy [14], [15]. Despite the performance improvement achieved through multimodal fusion, it requires well-matched optical-SAR image pairs both for training and inference, which limits its practical applications when only single-modal data are available on the real-running stage. In this article, we instead investigate the optical-aided SAR image feature learning method that only requires optical-SAR image pairs for pretraining and then adopt the pretrained model for SAR image classification. To do so, we propose a novel self-supervised contrastive learning technique that simultaneously optimizes the instance-level and cluster-level contrastive losses and design three manners to construct positive sample pairs for contrastive learning. To aid SAR feature learning by optical data, we generate the intermediate-level SAR images from the geolocation-matched optical data of an SAR image instance as its augmented "view." Our self-supervised network can be used as a pretrained model and be fine-tuned to achieve high classification performance with significantly less labeled data than the previous fully supervised methods.

The contributions of this article are summarized as follows.

1) We propose a novel self-supervised contrastive SAR feature learning framework that can utilize SAR images and their geolocation-matched optical data to pretrain SAR image feature extractor. Our framework performs instance-level and cluster-level contrastive learning pretext tasks, to further enhance semantic and discriminative representations of SAR images.
2) Moreover, we design three levels of positive pair construction strategies for contrastive learning: image self-augmentation, optical assistance, and graph searching. In addition to commonly used image self-augmentation, we generate intermediate SAR images from the geolocation-matched optical data as the augmented views for contrastive learning so that the pretrained model can assimilate the information from the optical data to enhance SAR features representation. We also introduce the graph search module to make full use of the global information of the training data and make the features more clustering friendly.
3) We conduct extensive experiments to demonstrate that our proposed method achieves state-of-art performance. By fine-tuning our model with a small number of labeled samples, the classification accuracy can quickly increase and surpass those of supervised methods on the SEN12MS dataset.

The remaining four sections of this article are organized as follows. Section II recaps the related work of SSL, deep clustering, and multimodal data utilization in RS. Section III presents the proposed SSL method optical-assistant (OA)-SSL. Experimental results and corresponding analysis are given in Section IV. Finally, Section V concludes this article.

II. Related Work

In this section, we present a brief overview of recent advancements that are related to our work: SSL, deep clustering, and multimodal RS image interpretation.

A. Self-Supervised Learning

In general, SSL approaches can generally be categorized into four groups depending on different pretext tasks: generation-based, context-based, free semantic label-based, and cross modal-based [4], [5]. The first two types of methods are more commonly used in the field of RS.

Generation-based methods in SSL are mainly to learn features by solving specific image generation tasks, such as image super-resolution [16], image inpainting [17], and image colorization [18]. For instance, masked image modeling endeavors to recover masked image patches from unmasked patches, with masked autoencoders (MAE [19]) being a prominent method for this purpose. On the other hand, context-based methods employ pretext tasks primarily based on the spatial structure and content similarity of images. By training the network to solve such tasks, these methods encourage the learning of transformation-invariant representations [20] or the capture of contextual and structural information in images [21]. In addition, in the field of RS, there are some data characteristics and prior information that can be utilized. For example, SeCo [22] utilizes temporal (seasonal) information to generate positive pairs under the contrastive learning framework, and Tile2Vec [23] adopts triplet-loss and selects close anchors as positive samples and far ones as negative samples to achieve unsupervised representation learning based on scene information.

B. Deep Clustering

Clustering is a fundamental unsupervised learning task that categorizes data into distinct groups. However, the performance of clustering heavily relies on the input features and their distribution structure in the feature space [6], [7]. Handcrafted features often exhibit weak discrimination and result in inferior clustering results, especially on large-scale complex datasets. Recent work integrates deep feature learning and clustering, which alternatively optimizes clustering on the learned representations and trains the feature extraction network using the cluster assignments as pseudolabels [6], [24]. However, the supervision signal only from cluster assignments is too weak to extract effective semantic features and often results in degenerate solutions [25]. Invariant information clustering (IIC) [25] aims to learn representations that are invariant to different versions of the same object by maximizing mutual information between their clustering assignments.

Some improved works combine contrastive learning and clustering that jointly perform contrastive learning both in feature and clustering space [26]. Contrastive clustering (CC) [7]

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conducts not only the instance-level but also the cluster-level contrastive learning, favoring clustering representations. Graph CC (GCC) [8] proposed a graph Laplacian-based CC loss to improve feature discriminative and clustering assignments.

C. Multimodal RS Image Interpretation

Nowadays, considerable multimodal RS images have become readily available, acquired by satellite platforms equipped with diverse imaging sensors. As a result, there is an increasing interest in leveraging such multimodal data for various interpretation tasks to enhance the robustness and reliability of decision-making by integrating complementary information [14].

Two typical methods for processing multimodal data are domain adaptation (DA) and data fusion (DF). DA addresses the challenge of model generalization across different domains. Its primary objective is to develop transfer techniques that enable trained models from a source domain to perform well on a different yet related target domain [27]. The general approach to achieving DA is to align features across different domains and find domain-invariant features. Existing methods can be roughly categorized into three groups: discrepancy-based methods [28], adversarial discriminative methods [29], [30], and adversarial generative methods [31]. Recently, Zhao et al. [32] proposed a feature decomposition-based adversarial domain adaption method to generalize ship detectors across different SAR platforms. Li et al. [33] proposed an adversarial domain adaption method for heterogeneous optical and SAR image classification.

DF aims to improve the model's performance by integrating complementary information extracted from multimodal data [14]. Various fusion strategies have been proposed for handling multimodal RS image data [34], [35]. Despite their success in improving the performance of decision-making, these methods need multimodal data for both training and testing. However, in some practical scenarios, it is highly likely that some modality data may be absent. In this article, we consider a practical scenario in which a number of SAR-optical data pairs are ready at hand to train a model, while only single-modality SAR data are available at the real testing or inference phase.

To tackle this practical problem, we investigate optical-assistant SAR feature learning, which only requires geo-matched optical-SAR image pairs to pretrain an SAR image feature backbone model. The pretrained SAR feature extractor is then fine-tuned with only SAR images, overcoming the limitations of the need for paired data in the inference stage with better flexibility and scalability.

III. Methodology

In this section, we elaborate on the proposed OA-SSL for SAR image classification that incorporates graph contrastive learning and leverages optical images to assist the learning of SAR image features. The overview of OA-SSL is provided in Section III-A, followed by a detailed introduction to its main modules. Sections III-B–III-D focus on the construction of positive pairs, while Sections III-E and III-F describe the design of the self-supervised pretext task. Section III-F provides a discussion on the total loss function and the training process.

A. Framework of the Proposed OA-SSL Method

The overall architecture of our proposed OA-SSL method, which employs the contrastive learning paradigm, is depicted in Fig. 1. Contrastive learning is to learn image representation by maximizing the similarity of the positive pairs while minimizing that of negative pairs. It consists of two key elements: 1) positive pair construction and 2) contrastive loss. One typical method uses augmented versions of the same image as positive pairs, while negative pairs come from a different image, achieving some invariant representation of transformations. Contrastive learning actually employs an instance-discrimination task that treats each image (or instance) as its own class and learns the representation to capture some invariance to image transformations. Our self-supervised SAR image feature learning framework consists of two key modules: the construction of positive image pairs, which relate to the features extraction from multimodal data, and the design of self-supervised clustering-style pretext tasks.

The strategies for generating positive SAR image pairs are illustrated in the gray box in Fig. 1. The construction of positive pairs is conducted at three levels by considering similarities in low-level image itself, middle-level optical assistance, and high-level global graph searching. At the low level, we use various image augmentations to construct positive pairs, just as most SSL methods do. At the middle level, we incorporate geographically matched optical images to generate SAR images. These geolocalization-matched optical images can be obtained more easily than the precise pixel-matched ones required by DF approaches. The rationale for this approach is supported by the observation that professional experts usually utilize optical images to aid their interpretation of SAR images. Therefore, optical images are utilized for pretraining the SAR image feature model, while only SAR images are required during the fine-tuning and testing phases. For high-level positive pairs, we involve a k-nearest neighbor (KNN) search for the training dataset to find some SAR samples with the highest feature similarity and use the corresponding generated SAR images as positive samples.

Effective pretext tasks play a crucial role in SSL. To improve feature representations and clustering, we have devised two contrastive SSL tasks at the instance and cluster levels, respectively.

B. Image Augmentation: Low-Level Positive Pairs

Low-level positive pairs are constructed through image augmentation techniques. For land cover classification, the semantic information of the image remains unchanged after certain geometric transformations. Given an SAR image I, we randomly select two image transformations, T_a and T_b, where the transformed images $\{I_a, I_b\} = \{T_a(I), T_b(I)\}$ are considered positive samples. As is well-known, SAR data is influenced by the geometric characteristics of the scene as

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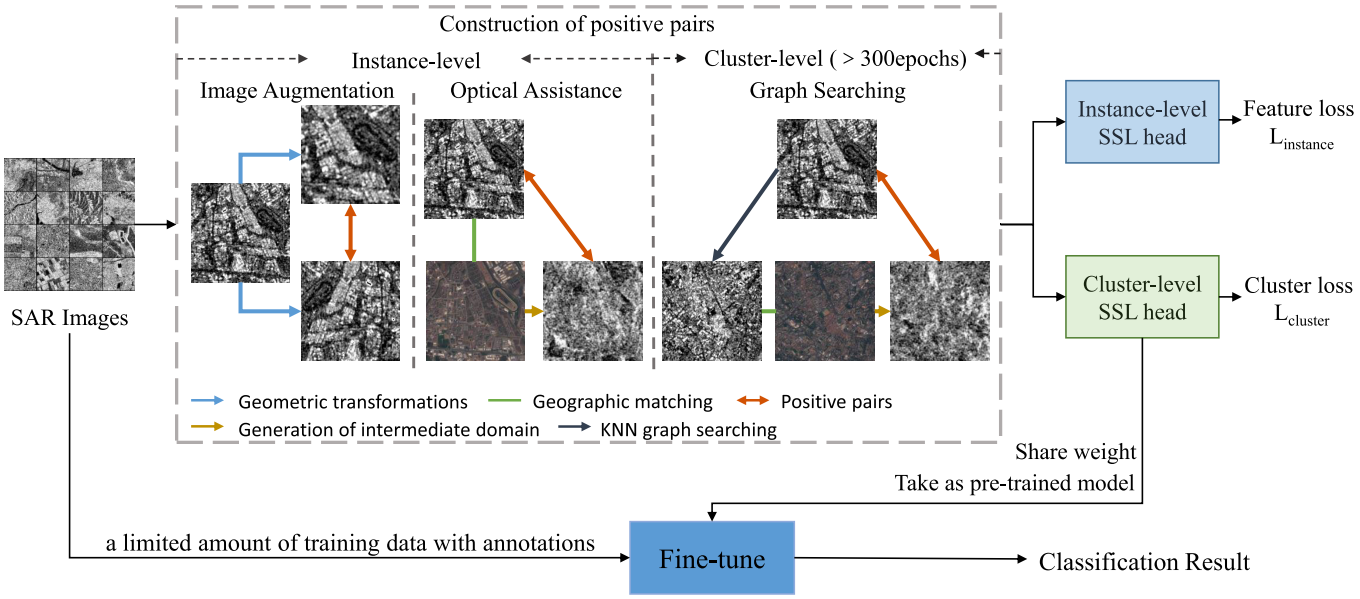


Fig. 1. Proposed OA-SSL framework for SAR image classification consists of two main steps: pretraining with a large set of unlabeled samples and subsequent fine-tuning. Self-supervised signals are generated using contrastive learning, where positive sample pairs are constructed at three different levels: low-level through image augmentation, mid-level with optical assistance, and high-level using graph searching. Concurrently, negative pairs are generated from other samples within each minibatch. To further enhance feature extraction and enable effective classification, instance-level, and cluster-level SSL heads are incorporated into the framework.

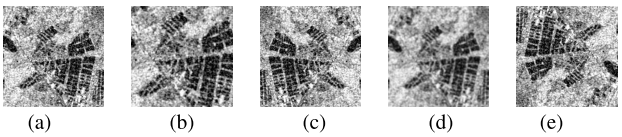


Fig. 2. Image augmentation examples. (a) Original. (b) Crop. (c) Flip. (d) Blur. (e) Rotation.

well as the position and orientation of the sensor. However, to simplify the problem, we utilize a collection of widely adopted image augmentation techniques [4], such as random resized crop, horizontal flip, blur, and rotation, to enhance the diversity and variety of the dataset. Fig. 2 showcases several examples of the augmented images.

C. Optical Assistance: Middle-Level Positive Pairs

The construction of middle-level positive pairs draws inspiration from the process of human annotating SAR images, usually aided by optical images. This assistance is particularly valuable for low- or medium-resolution SAR images, such as those from the Sentinel-1A sensor. During the human annotation process, image interpreters identify similar semantic features between SAR images and their geomatched optical images, and employ these features to categorize SAR contents. This annotation process exhibits several distinct characteristics: 1) human annotators rely on semantic features, not image pixels, for matching; 2) the semantic features typically relate to edges, intersections, and topological structures, rather than image density or texture; 3) geographical-matched, rather than precise pixelwise matched, optical images suffice for complementing SAR data. The process is further elucidated in Fig. 3. Consequently, geographically matched optical data can be employed to generate SAR positive pairs for contrastive learning.

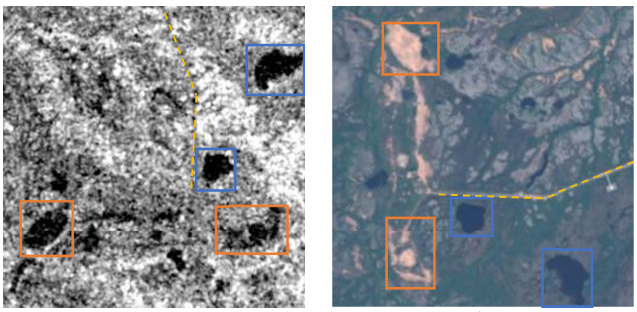


Fig. 3. Semantic alignment of SAR-optical image pairs. By comparing key textures such as landmarks and river trends, SAR images and optical images can be semantically aligned. With the help of optical images, some features that will be confused with SAR single-source images can be better distinguished. For example, the blue box of water and the orange box of barren, although both appear as dark areas in SAR images, have significant differences in optical images. (a) SAR image. (b) Optical image.

In practice, it is easy to obtain geographically matched SAR-optical image pairs with some toolboxes like Google Earth. However, the essential differences in imaging mechanisms between SAR and optical sensors lead to domain-specific low-level features. These differences pose a challenge to close the gap between the two domains. To bridge this gap between domains, we utilize the CycleGAN framework [36] to generate fake SAR data from optical data. Nevertheless, the conventional CycleGAN may not yield satisfactory SAR data for effective contrastive learning. To address this, we improve CycleGAN by introducing a novel loss term that enforces semantic feature alignment, as depicted in Fig. 4.

The modified CycleGAN comprises two symmetrical components: an encoder E_S, a generator G_S, and a discriminator D_S in the SAR domain X, as well as their counterparts E_O, G_O, and D_O, respectively, in the optical domain Y. Given an

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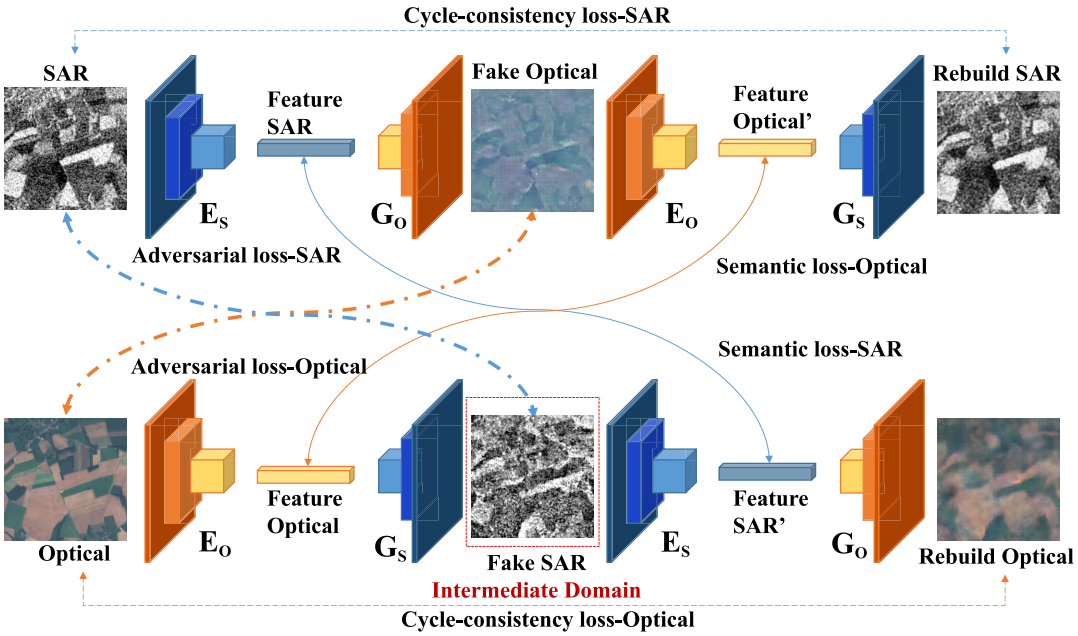


Fig. 4. Adversarial learning framework for aligning SAR and optical features and generating fake-SAR images for positive pair construction at the middle level. The generation process is guided by three loss functions: adversarial discrimination loss, cycle consistency loss, and feature semantics loss. The goal is to retain the optical semantic information while transferring the underlying feature representations to the SAR domain, thereby facilitating feature extraction with shared parameters.

SAR image $x \in X$, we encode it as $E_S(x)$ and generate a fake optical image $y_{\text{fake}} = G_O(E_S(x))$. Subsequently, y_{fake} is passed through E_O to obtain $E_O(y_{\text{fake}})$, which is then used to reconstruct the original SAR image as $x_r = G_S(E_O(y_{\text{fake}}))$. Similarly, for an optical image $y \in Y$, we generate a fake SAR image x_{fake} and a reconstruction y_r, with the corresponding features being $E_O(y)$ and $E_S(x_{\text{fake}})$. We denote the fake SAR and optical domains as $\widetilde{X}$ and $\widetilde{Y}$, respectively. The alignment of SAR and optical images is achieved by optimizing a loss function that consists of the following three components.

1) *Adversarial Loss (Loss-D):* It involves optimizing a game process, where the encoder and generator are trained to deceive the discriminator and produce images that closely resemble the target domain. The formulation is

$$L_D = \min_{E_O, G_S} \max_{D_S} \text{Loss}_D\left(X, \widetilde{X}\right) + \min_{E_S, G_O} \max_{D_O} \text{Loss}_D\left(Y, \widetilde{Y}\right) \quad (1)$$

where

$$\text{Loss}_D\left(X, \widetilde{X}\right) = \mathbb{E}_{x\sim X}\left[\log D_X(x)\right] + \mathbb{E}_{\widetilde{x}\sim \widetilde{X}}\left[\log\left(1 - D_X\left(\widetilde{x}\right)\right)\right] \quad (2)$$

and $\text{Loss}_D(Y, \widetilde{Y})$ is defined similarly.

2) *Cycle-Consistency Loss (Loss-C):* The cycle-consistency loss directly compares the original image and the reconstructed image per pixel as

$$\text{Loss}_C = \mathbb{E}_{x\sim X}\left[\|x_r - x\|_1\right] + \mathbb{E}_{y\sim Y}\left[\|y_r - y\|_1\right]. \quad (3)$$

3) *Semantic Loss (Loss-F):* This loss function strengthens the traditional CycleGAN by imposing semantic constraints into the generative process. It aims to minimize semantic information loss during domain transfer by comparing the features extracted from the generated image with the features of the original image using cosine distance loss

$$\text{Loss}_F = \mathbb{E}_{x\sim X}[d\langle E_S(x_{\text{fake}}), E_S(x)\rangle] + \mathbb{E}_{y\sim Y}\left[d\langle E_O(y_{\text{fake}}), E_O(y)\rangle\right]. \quad (4)$$

After generating fake SAR images from geolocation-matched optical data, we create positive image pairs by combining them with the original SAR images.

D. Graph Searching: High-Level Positive Pairs

Low-level and mid-level positive pairs are constructed from the image itself and multimodal data. Additionally, we leverage the global information of the whole dataset to construct positive pairs to increase clustering stability.

To achieve this, we employ an additional memory bank [37] to store computed features. Feature similarity is then calculated to construct a KNN graph, as depicted in Fig. 5. The underlying concept is that a sample and its KNNs should belong to the same cluster, and these samples, along with their augmented counterparts, can be paired as positive samples.

E. Instance-Level Contrastive SSL Head and Loss

Contrastive learning compares each image (or "instance") and its augmented version by image transforms as an instance discrimination task, yielding some invariant features to image transformations [7], [37].

Given a mini-batch of samples $\{x_1, x_2, \ldots, x_N\}$ of size N, we perform one of our three positive-pair construction methods, obtaining $2N$ data augmentation samples

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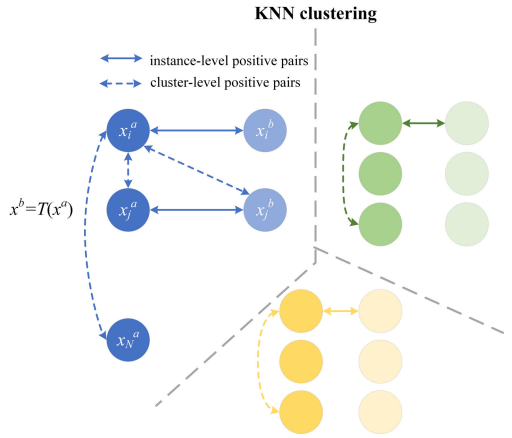


Fig. 5. Construction of high-level positive pairs by graph searching. This process involves calculating and updating the features of all samples in the training set, resulting in the establishment of a KNN graph. It is assumed that samples and their KNNs belong to the same cluster and can be paired as positive samples. Graph searching utilizes the entire training set and contributes to learning more discriminative and clustering-friendly features.

$\{x_1^a, x_2^a, \ldots, x_N^a, x_1^b, x_2^b, \ldots, x_N^b\}$. For a specific sample x_i, there is one positive pair, (x_i^a, x_i^b), and $2N-2$ negative pairs $(x_i^a, x_j^{a(b)})$, $j \neq i$. The goal of contrastive learning is to push positive sample pairs closer to each other while pulling away negative sample pairs. The instance-level SSL head is implemented as a two-layer MLP, which maps the features extracted by the backbone to a 128-D feature, as shown in Fig. 6. Let $\mathbf{f}_i^a$ and $\mathbf{f}_i^b$ be the extracted features of x_i^a and x_i^b, respectively, and the pairwise similarity between the image instances is measured by the cosine distance

$$s\left(\mathbf{f}_i^a, \mathbf{f}_j^b\right) = \frac{\left(\mathbf{f}_i^a\right)\left(\mathbf{f}_j^b\right)^T}{\left\|\mathbf{f}_i^a\right\|_2 \left\|\mathbf{f}_j^b\right\|_2}. \tag{5}$$

Referring to CC [7], the loss function for a specific sample is designed as

$$\ell_i^a = -\log \frac{\exp\left(s\left(\mathbf{f}_i^a, \mathbf{f}_j^b\right)/\tau_I\right)}{\sum_{j=1}^{N}\left[\exp\left(s\left(\mathbf{f}_i^a, \mathbf{f}_j^a\right)/\tau_I\right) + \exp\left(s\left(\mathbf{f}_i^a, \mathbf{f}_j^b\right)/\tau_I\right)\right]} \tag{6}$$

where τ_I is the instance-level temperature parameter. And the total instance-level contrastive loss is computed over every augmented sample and three types of positive pairs

$$\mathcal{L}_{\text{ins}} = \frac{1}{2N}\sum_{i=1}^{N}\left(\ell_i^a + \ell_i^b\right) \tag{7}$$

$$\mathcal{L}_{\text{ins}} = \mathcal{L}_{\text{ins}}^{\text{Low}} + \mathcal{L}_{\text{ins}}^{\text{Mid}} + \mathcal{L}_{\text{ins}}^{\text{High}}. \tag{8}$$

F. Cluster-Level Contrastive SSL Head and Loss

In addition to instance-level comparison, we incorporate a cluster-level discriminative task that compares clustering representation to maximize the distance between clusters, helping to learn more discriminative and clustering-friendly features [7], [8].

By adopting a multilayer MLP as a classifier, a probability vector will be generated with the extracted features input, where each dimension of the vector represents the probability of being assigned to the corresponding class. Considering the matrix integrated by the entire batch probability vector, the column vector can be used as a feature of the clustering categories, indicating which samples are included in the category [7], as shown in Fig. 6.

Suppose clustering into C categories and getting $2C$ class features $\{\mathbf{y}_1^a, \mathbf{y}_2^a, \ldots, \mathbf{y}_C^a, \mathbf{y}_1^b, \mathbf{y}_2^b, \ldots, \mathbf{y}_C^b\}$ after two transformations, where $\mathbf{y}_i$ is an N-dimensional vector. Then, $(\mathbf{y}_i^a, \mathbf{y}_i^b)$ can form a positive pair and the other $2C-2$ clusters form negative pairs. The cluster-level pretext task can increase the class separability and discriminability of the extracted features by maximizing positive pairs consistency while minimizing negative pairs similarity, making them more stable and clustering friendly.

The loss function for a specific cluster is designed as

$$\hat{\ell}_i^a = -\log \frac{\exp\left(\widetilde{s}\left(\mathbf{y}_i^a, \mathbf{y}_i^b\right)/\tau_C\right)}{\sum_{j=1}^{C}\left[\exp\left(\widetilde{s}\left(\mathbf{y}_i^a, \mathbf{y}_j^a\right)/\tau_C\right) + \exp\left(\widetilde{s}\left(\mathbf{y}_i^a, \mathbf{y}_j^b\right)/\tau_C\right)\right]} \tag{9}$$

where $\widetilde{s}(\mathbf{y}_i^a, \mathbf{y}_i^b) = s((\mathbf{y}_i^a)^T, (\mathbf{y}_i^b)^T)$, and τ_C is the cluster-level temperature parameter [7]. And the total cluster-level contrastive loss is

$$\mathcal{L}_{\text{cluster}} = \frac{1}{2C}\sum_{i=1}^{C}\left(\hat{\ell}_i^a + \hat{\ell}_i^b\right) \tag{10}$$

$$\mathcal{L}_{\text{cluster}} = \mathcal{L}_{\text{cluster}}^{\text{Low}} + \mathcal{L}_{\text{cluster}}^{\text{Mid}} + \mathcal{L}_{\text{cluster}}^{\text{High}} - H(Y) \tag{11}$$

where $H(Y)$ is the entropy of cluster assignment probabilities, which helps to avoid the trivial solution that most instances are assigned to the same cluster.

G. Total Cost Function and Training

The two SSL losses are simultaneously optimized, and the total cost function is

$$\mathcal{L} = \mathcal{L}_{\text{ins}} + \alpha\mathcal{L}_{\text{cluster}} \tag{12}$$

where the constant α balances the instance-level loss $\mathcal{L}_{\text{ins}}$ and the cluster-level loss $\mathcal{L}_{\text{cluster}}$. Here, we empirically set α to 1, indicating equal importance of $\mathcal{L}_{\text{ins}}$ and $\mathcal{L}_{\text{cluster}}$. This parameter setting is further validated by the ablation study on pretext SSL tasks. In this study, our proposed approach demonstrates comparable performance when trained independently with either the instance-level or cluster-level tasks.

To self-supervisedly pretrain the SAR backbone model $\mathcal{F}$ aided by the optical data for SAR image classification, we firstly train the generative model of intermediate-level SAR images $\mathcal{M}_{o\to s}$ using the geolocation-matched optical-SAR pairs dataset via the method introduced in Section III-C. A large number of intermediate-level SAR images are then generated from the geolocalization-matched optical images and stored to construct the corresponding positive pairs for contrastive feature learning.

Besides directly using the generated SAR images as the positive samples, we also perform KNN retrieval on the original SAR images during the backbone model training in the feature space extracted and use their corresponding generated images to construct graph-matched positive pairs.

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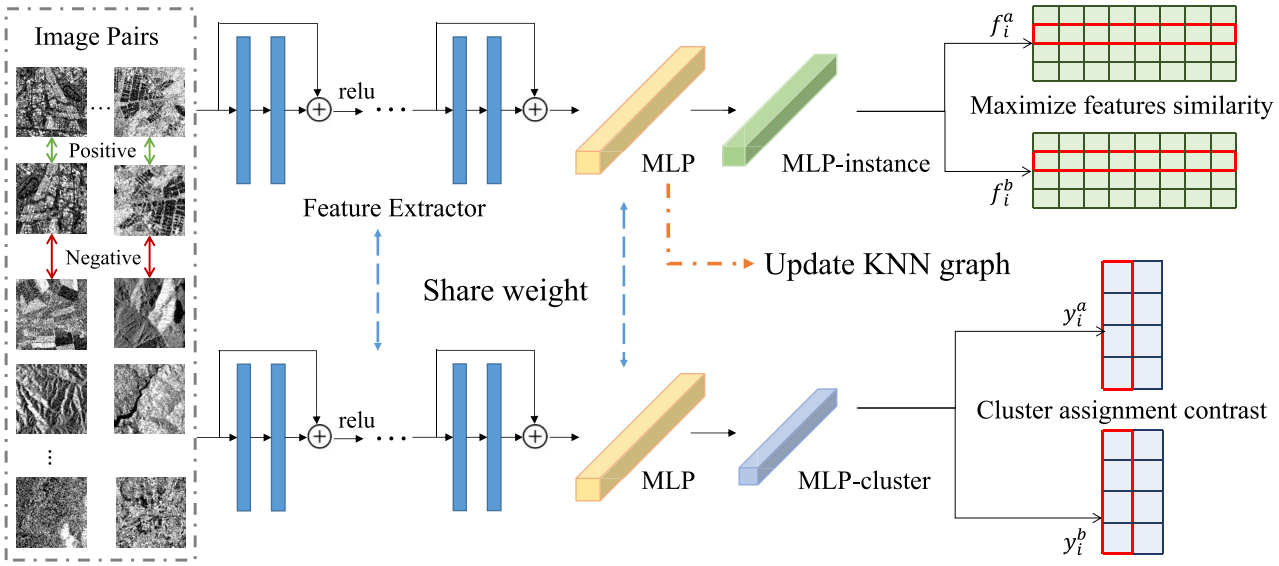


Fig. 6. Instance-level and cluster-level contrastive SSL pretext tasks. Popular feature extraction backbones, such as ResNet, VGG, inception, and transformer, are used to extract image features. The KNN graph of features is updated at every m epoch (searching unit). Two independent MLP networks are designed, one for further feature extraction and the other for clustering. For instance samples, positive sample pairs should have higher feature similarity. For clusters, two groups of positive samples should have similar category distributions and the other clusters should be much more discriminative.

To make the training stable and reduce the computational burden, we conduct a KNN search every m epoch and use the average features of the m epochs to construct the KNN graph during the backbone model training.

The training and testing procedure of the SAR backbone model is summarized in Algorithm 1.

IV. Experimental Results and Analysis

In this section, we present in detail the experimental datasets, the training settings, and the experimental results. Furthermore, a comparison experiment with the existing SOTA self-supervised image classification methods and a series of ablation studies are conducted to demonstrate the effectiveness and superiority of our proposed method.

A. Dataset

Experiments were conducted on the publicly available dataset SEN12MS [38], which comprises triplets of the Sentinel-1 SAR images, multispectral Sentinel-2 images, and MODIS land cover maps. The resolution of the SAR image is 20 m per pixel, and the optical image is 10 m per pixel. The patches are distributed across the lands of the Earth and over all four meteorological seasons. SAR images and optical images with the same serial number have been geographically registered.

In the experiments, we use images with the summer label and choose the Sentinel-1 VH polarization channel and the Sentinel-2 RGB channels.

The first channel of MODIS land cover and the International Geosphere and Biosphere Program (IGBP) classification scheme is used to label the images. The category that accounts for more than 75% of pixels of a patch is set to its label. After we filter out the categories with too few samples and merge similar categories, the images are divided into eight categories: barren, croplands, forests, grasslands, savannas, shrublands, urban, and water, for a total of 28.5k patches. The details of our processed SEN12MS dataset are shown in the Table I. Examples of the dataset are shown in the first column of Fig. 7.

TABLE I
Details of Our Processed SEN12MS Dataset

Class	IGBP-Label	Patches	Train
barren	16	751	396
croplands	12, 14	6,464	3,182
forests	1, 2, 3, 4, 5	3,252	1,593
grasslands	10	4,105	2,038
savannas	8, 9	3,897	1,996
shrublands	6, 7	4,490	2,227
urban	13	1,944	963
water	17	3,656	1,813
Total	-	28,559	14,208

B. Training Details

Our model is based on the PyTorch framework. The Adam optimizer is adopted to train the model, with a l_r of 0.0003 and a weight decay of $1e-4$. The batch size has a direct impact on the proportion of positive and negative pairs and is set to 128 according to experimental comparisons. For KNN graph searching, the searching unit m is set to 5, and k is set to 3. The temperature parameters τ are employed to regulate the attention degree toward negative samples. In the case of the instance-SSL task, we need to pay high attention to similar negative samples to prevent model collapse and improve feature learning ability. Therefore, we set a smaller value for τ. However, some samples in the same mini-batch may belong to the same category and have high similarity. In order to avoid the model paying too much attention to these difficult negative samples, the temperature coefficient

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Algorithm 1 OA-SSL

Input: Training dataset X, training epochs E, batch size N, cluster number C, initial feature extractor F, instance-level MLP g_I, and cluster-level MLP g_C.

Output: pretrained SAR backbone feature extractor $\mathcal{F}$

% *Intermediate-level SAR images generation*
Train the intermediate SAR generation model $\mathcal{M}$ using the dataset of geolocation-matched optical-SAR pairs.
Generate intermediate-level SAR images from the geolocation-matched optical images of the original SAR images with the model. $\mathcal{M}$.

% *SAR backbone feature extractor $\mathcal{F}$ training*
Initializing the model $\mathcal{F}$
for *epoch* $= 1$ to E **do**
  Sampling a mini-batch $\{x_i\}_{i=1}^{N}$ from X.
  Extract and record the features $\mathbf{f_i} = \mathcal{F}(x_i)$.
  Update the KNN graph every 5 epochs.

  Generate positive pairs using three methods: data augmentation transformation ($\mathcal{T}$), optical assistance matching ($\mathcal{M}$), and KNN graph searching ($\mathcal{S}$):
  A: data augmentation by random low-level image transformation$\{\mathcal{T}^a(x_i), \mathcal{T}^b(x_i)\}$,
  O: find generated intermediate-level SAR images from the geolocalization-matched optical images$\{\mathcal{T}^a(x_i), \mathcal{T}^b[\mathcal{M}(x_i)]\}$,
  S: find the corresponding generated SAR images of the KNN searched SAR images $\{\mathcal{T}^a(x_i), \mathcal{T}^b\{\mathcal{M}[\mathcal{S}(x_i)]\}\}$.

  Compute instance-level contrastive loss in Eqn. (6)-(8).

  Compute cluster-level contrastive loss in Eqn. (9)-(11).

  Optimize the model F, g_I, and g_C using the Adam optimizer by minimizing the overall loss in Eqn. (12).
end for

% Compute clustering assignments
Extract features by $\mathbf{f} = \mathcal{F}(x)$ with input x.
Compute cluster assignment by $c = argmax\{g_C(\mathbf{f})\}$.

% Fine-tuning only with a limited number of labeled SAR data $\{x_i^{SAR}, y_i^{SAR}\}$
Add a linear classifier layer on the top of the pretrained network.
Supervised training the whole network initialized from the pretrained model
Or Supervised training the linear classification layer while freezing the pretrained model.

in the instance level SSL τ_I is taken as 0.5. Clustering aims to increase category separability and has no preference for specific classes, with τ_C of 1. All experiments are performed on a workstation with NVIDIA QUADRO RTX 6000 GPUs and CUDA 11.4 under Ubuntu 18.04.

C. Generation Results in Intermediate Domain

In Fig. 7, we present a set of randomly sampled images obtained using the proposed adversarial learning network for constructing middle-level positive pairs. The first column showcases the original SAR and optical images, which are geographically matched. The second column displays the generated fake-SAR and fake-optical images, where the fake-SAR images are employed for constructing middle-level positive pairs. The third column shows the reconstructed SAR and optical images.

We can observe that the reconstructed SAR and optical images closely resemble the original counterparts, indicating the successful convergence of the proposed adversarial learning network. While the fake-SAR and fake-optical images appear visually distinct from the originals, we believe that they still retain crucial semantic features that can contribute to SAR image classification. Further analysis is provided in the following.

1) Images generated by the generative module retain most of the semantic information. From the reconstructed optical image in Fig. 7(f), we can find the reconstructed optical images roughly restore the correct color and texture structure, even though the images in the intermediate domain are grayscale without color appearance. And by comparing Fig. 7(a) and (c), the reconstruction loss of SAR images is very low. This verifies that the generation module operates well, and the generated images have encoded the useful semantic information.
2) Fake-SAR images and SAR images exhibit different spatial patterns. The appearance of the generated fake-SAR images is not only close to real SAR images but also retains some characteristics of optical images (texture, structure, and so on). This cross-domain image generation can be regarded as a special kind of data augmentation to enhance SAR feature extraction.

By bridging the optical and SAR through an intermediate domain, the intermediate images can retain more semantic information shared by SAR and optical images while ignoring domain-specific information. It is conducive to the self-supervised extraction of more effective features and classification.

D. Comparison Experiments

In this section, we compare our proposed OA-SSL method to supervised methods, self-supervised methods, DA methods, and DF methods. The training data requirements of these methods are depicted in Fig. 8. Supervised methods necessitate a large amount of labeled SAR data to train the model from scratch. Self-supervised methods utilize unlabeled SAR data for pretraining the model and fine-tune it with a small amount of labeled SAR data, thereby reducing data requirements. DA methods classify SAR data using models transferred from those trained on optical data, thus requiring a significant amount of labeled optical data and unlabeled SAR data.

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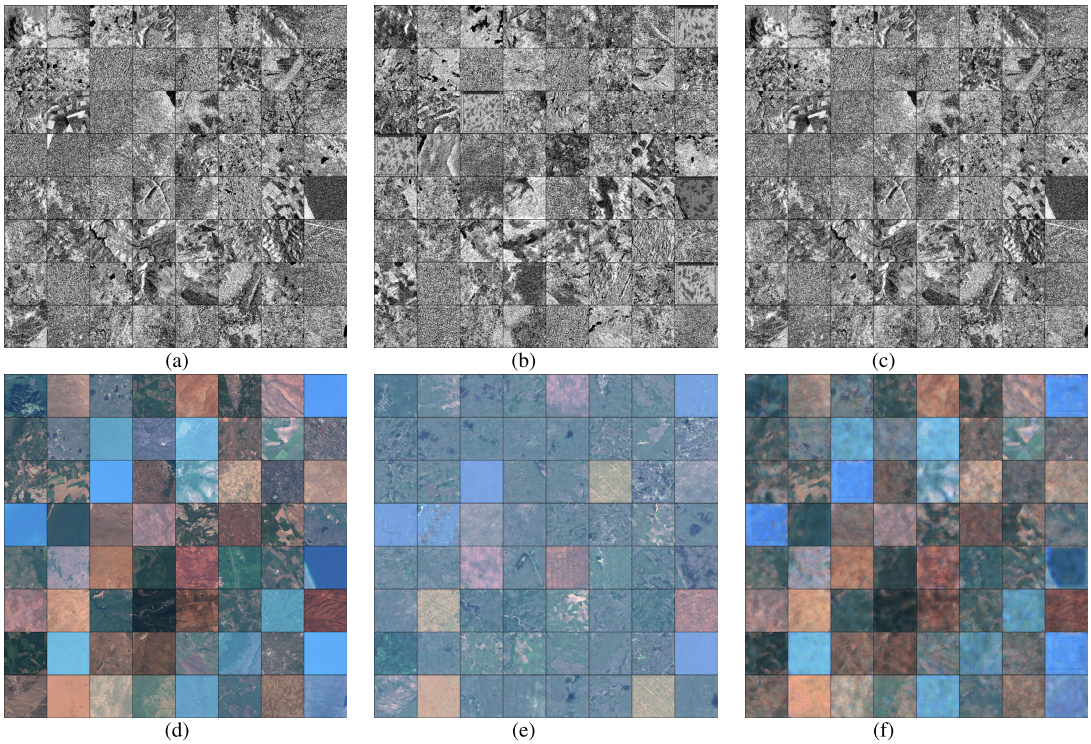


Fig. 7. Samples generated by our adversarial learning method. From left to right: the original image, the generated fake image, and the reconstructed image. The top row corresponds to the SAR domain, while the bottom row corresponds to the optical domain. The generated "fake SAR" images serve as the intermediate domain, which implicitly preserves most of the original optical image features, but has been migrated to the feature space consistent with the SAR domain. (a) SAR. (b) Fake-SAR. (c) Rebuilt-SAR. (d) Optical. (e) Fake-optical. (f) Rebuilt-optical.

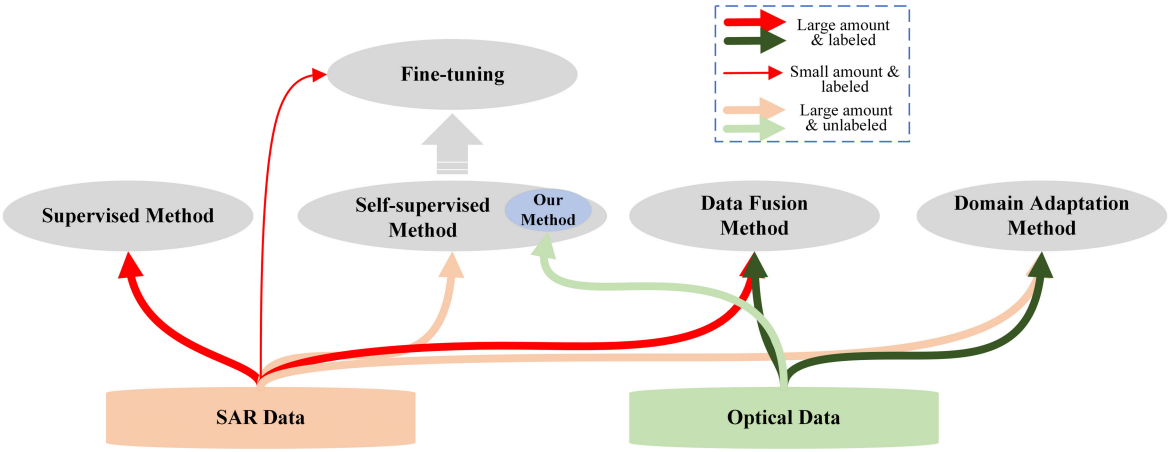


Fig. 8. Data requirements of different methods. Our proposed OA-SSL method only requires unlabeled SAR and optical image pairs that are geolocation-matched to each other during the pretraining stage, as well as a small amount of labeled SAR data during the fine-tuning stage.

While DA methods reduce the need for labeled SAR data, their performance is lower compared with the other methods mentioned earlier. DF methods aim to leverage complementary information between SAR and optical data to enhance performance. However, they require precisely matched SAR and optical image pairs for training and model application, limiting their practical applicability.

To comprehensively evaluate our OA-SSL method, we have selected six classical self-supervised methods: CC [7], GCC [8], MoCo [39], Deepcluster [6], IIC [25], and MAE [19], along with two DA methods, DANN [29] and CDAN [30], and one DF method, SatViT [40], for comparison. Additionally, we have conducted comparative experiments using various feature extraction backbones, including ResNet18 [41], ResNet50, VGG16 [42], Inception_v3 [43], and Swin-Transformer Tiny [44]. Five widely used metrics: overall accuracy (OA), average accuracy (AA), average precision (AP), F1 score (F1), and kappa coefficient (Kappa), are used to evaluate the classification performance of methods. Normalized mutual information (NMI) and adjusted rand index (ARI) are used to evaluate the clustering performance of self-supervised methods. Higher values for these metrics indicate a more accurate classification. We randomly split the dataset into a training set and a test set in a 1:1 ratio.

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TABLE II
CLASSIFICATION RESULT ON SEN12MS

BackBone	Method	Setting	OA	AA	AP	F1	Kappa	NMI	ARI
ResNet18	DANN [29]	DA	0.2527	0.2706	0.2731	0.1944	0.1329	-	-
	CDAN [30]		0.2377	0.2639	0.2527	0.1688	0.1163	-	-
	Deepcluster [6]	SSL	0.3290	0.3191	0.3233	0.3080	0.2316	0.2070	0.1335
	IIC [25]		0.3406	0.3541	0.3561	0.3358	0.2441	0.1994	0.1211
	MoCo [39]		0.3253	0.3527	0.3282	0.3156	0.2272	0.1780	0.1011
	CC [7]		0.3748	0.3917	0.3766	0.3738	0.2794	0.2513	0.1575
	GCC [8]		0.3780	0.4060	0.3752	0.3763	0.2867	0.2656	0.1606
	OA-SSL (our)		**0.5608**	**0.6320**	**0.5610**	**0.5500**	**0.4983**	**0.5155**	**0.3793**
	Train from Scratch	FT	0.8886	0.8815	0.8915	0.8844	0.8686	-	-
	OA-SSL-FT (our)		0.9167	0.9159	0.9165	0.9159	0.9019	-	-
	Pretrained with ImageNet		0.9223	0.9206	0.9082	0.9093	0.9088	-	-
	Concat-4 channel	DF	0.9350	0.9317	0.9470	0.9383	0.9234	-	-
ResNet50	DANN [29]	DA	0.2499	0.2761	0.3100	0.1856	0.1350	-	-
	CDAN [30]		0.2218	0.2469	0.2352	0.1583	0.1031	-	-
	Deepcluster [6]	SSL	0.3503	0.3537	0.3170	0.3257	0.2465	0.2233	0.1438
	IIC [25]		0.3536	0.3748	0.3458	0.3459	0.2580	0.2217	0.1456
	MoCo [39]		0.3633	0.3194	0.3144	0.2867	0.2392	0.2235	0.1193
	CC [7]		0.3859	0.3959	0.3780	0.3790	0.2920	0.2694	0.1683
	GCC [8]		0.3755	0.4001	0.3722	0.3721	0.2845	0.2670	0.1602
	OA-SSL (our)		**0.5379**	**0.6109**	**0.5345**	**0.5307**	**0.4701**	**0.4923**	**0.3667**
	Train from Scratch	FT	0.8960	0.8802	0.9003	0.8865	0.8776	-	-
	OA-SSL-FT (our)		0.9202	0.9077	0.9213	0.9135	0.9060	-	-
	Pretrained with ImageNet		0.9192	0.8300	0.8153	0.8204	0.9047	-	-
	Concat-4 channel	DF	0.9160	0.9187	0.9210	0.9176	0.9012	-	-
VGG16	DANN [29]	DA	0.3032	0.2583	0.2159	0.1775	0.1287	-	-
	CDAN [30]		0.2713	0.2040	0.1994	0.1362	0.0666	-	-
	Deepcluster [6]	SSL	0.3693	0.3836	0.3893	0.3779	0.2697	0.2534	0.1471
	IIC [25]		0.3650	0.3780	0.3530	0.3544	0.2682	0.2352	0.1584
	MoCo [39]		0.3444	0.3929	0.3452	0.3488	0.2464	0.2003	0.1274
	CC [7]		0.4221	0.4389	0.4166	0.4099	0.3374	0.3094	0.2101
	GCC [8]		0.4246	0.4400	0.4247	0.4145	0.3418	0.3230	0.2185
	OA-SSL (our)		**0.5442**	**0.6236**	**0.5523**	**0.5385**	**0.4791**	**0.5315**	**0.3760**
	Train from Scratch	FT	0.9028	0.9038	0.9153	0.9091	0.8854	-	-
	OA-SSL-FT (our)		0.9424	0.9407	0.9498	0.9449	0.9320	-	-
	Pretrained with ImageNet		0.9328	0.9383	0.9207	0.9292	0.9207	-	-
	Concat-4 channel	DF	0.9213	0.9227	0.9303	0.9263	0.9074	-	-
Inception_v3	DANN [29]	DA	0.2196	0.2170	0.3148	0.1585	0.0941	-	-
	CDAN [30]		0.1973	0.2014	0.3048	0.1553	0.070	-	-
	Deepcluster [6]	SSL	0.3538	0.3774	0.3779	0.3624	0.2592	0.2201	0.1309
	IIC [25]		0.3507	0.3734	0.3650	0.3540	0.2558	0.2328	0.1417
	MoCo [39]		0.3332	0.2336	0.1663	0.1564	0.1637	0.1785	0.0827
	CC [7]		0.4073	0.4114	0.4422	0.3739	0.3239	0.3078	0.2013
	GCC [8]		0.4552	0.5285	0.4535	0.4526	0.3758	0.3972	0.2639
	OA-SSL (our)		**0.6027**	**0.5690**	**0.6015**	**0.5697**	**0.5458**	**0.5781**	**0.4372**
	Train from Scratch	FT	0.9214	0.8676	0.9280	0.8836	0.9072	-	-
	OA-SSL-FT (our)		0.9804	0.9811	0.9817	0.9813	0.9770	-	-
	Pretrained with ImageNet		0.9521	0.9465	0.9524	0.9486	0.9437	-	-
	Concat-4 channel	DF	0.9489	0.9558	0.9529	0.9540	0.9399	-	-
Swin-Transformer Tiny	MoCo [39]	SSL	0.3491	0.3548	0.3548	0.3358	0.2578	0.1810	0.1221
	CC [7]		0.4055	0.4291	0.4003	0.3971	0.3188	0.2935	0.1947
	GCC [8]		0.4356	0.4336	0.4369	0.4163	0.3560	0.3328	0.2275
	OA-SSL (our)		**0.5990**	**0.5677**	**0.6002**	**0.5693**	**0.5403**	**0.5395**	**0.4162**
	OA-SSL-FT (our)	FT	0.9572	0.9573	0.9619	0.9595	0.9497	-	-
	Pretrained with ImageNet		0.9184	0.8256	0.8073	0.8158	0.9035	-	-
ViT	MAE [19]	SSL	0.3913	0.3951	0.4189	0.3843	0.2944	0.2590	0.1557
	SaViT-4channel [40]	DF	0.9678	0.9705	0.9699	0.9702	0.9621	-	-

Bold: Optimal performance of self-supervised trained models Underline: Optimal performance of supervised trained models

For methods without a built-in clustering function, we apply k-means to the features extracted by the methods. The experimental results are shown in Table II.

Our self-supervised method outperforms a wide range of other SOTA self-supervised methods. After fine-tuning, it surpasses supervised baselines trained from scratch and remains competitive with DF methods.

Analyzing the significant performance differences among these methods, we find that the OA-SSL method, based on the self-supervised paradigm, can efficiently utilize features from both SAR and optical domains, demonstrating superior applicability and scalability. The significant domain shift between SAR and optical images poses a challenge for DA, as it primarily focuses on learning domain-invariant features, resulting in the loss of discriminative information and subpar classification performance. In contrast, DF methods leverage multimodal data and combine information from all domains to achieve optimal classification performance.

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TABLE III
ABLATION STUDIES—DESIGN OF POSITIVE PAIRS

Domains	Positive	Backbone	OA	AA	F1	Kappa	NMI
Optical	L+M+H	ResNet18	0.46	0.47	0.45	0.38	0.40
Gray	L+M+H		0.48	0.49	0.48	0.47	0.39
Generate	L		0.37	0.39	0.37	0.28	0.25
	L+M		0.54	0.58	0.53	0.46	0.44
	L+M+H		0.56	0.63	0.55	0.50	0.52
Optical	L+M+H	VGG16	0.46	0.52	0.46	0.38	0.39
Gray	L+M+H		0.48	0.48	0.49	0.47	0.41
Generate	L		0.42	0.44	0.41	0.34	0.31
	L+M		0.54	0.62	0.54	0.48	0.53
	L+M+H		0.55	0.63	0.54	0.48	0.53
Optical	L+M+H	Inception	0.49	0.52	0.46	0.42	0.37
Gray	L+M+H		0.50	0.48	0.47	0.42	0.37
Generate	L		0.41	0.41	0.37	0.32	0.31
	L+M		0.55	0.52	0.50	0.47	0.51
	L+M+H		0.60	0.57	0.57	0.55	0.58

However, their reliance on multimodal data for both training and testing limits their practicality and flexibility.

In comparison, our proposed method offers several advantages. First, it requires only weakly matched, unlabeled SAR-optical image pairs for training, which can be easily obtained. Second, SAR image classification is assisted by optical images, which implicitly integrate characteristics of both domains to facilitate classification. Third, during inference, only SAR images are needed without any additional information. Additionally, the model can be flexibly fine-tuned, making it highly promising for various applications.

E. Ablation Studies

In this part, we present a series of ablation experiments to thoroughly evaluate our proposed method. We conduct four ablation studies, focusing on the following aspects: 1) investigating the design of positive pairs; 2) analyzing the impact of pretext SSL tasks; 3) exploring the influence of the k parameter in KNN graph searching; and 4) examining the effect of different cluster number c.

1) Design of Positive Pairs: Our proposed intermediate domain is compared with the original optical images and grayscale ones, demonstrating its significant advantages and ability to improve classification performance. Ablation experiments are also conducted to evaluate the effectiveness of traditional data augmentation methods and the other two positive pair construction methods. Results are shown in Table III. Additionally, the learned features are visualized in Fig. 9, which illustrates that employing the generated image from optics to SAR as the intermediate domain effectively enhances feature extraction, resulting in stronger class separability and facilitating subsequent fine-tuning.

2) Impact of the Pretext SSL Tasks: To assess the impact of the two pretext tasks, we compare the performance of SSL and fine-tuned models trained with a single task against those trained with a combination of both tasks. For the sake of experimental stability, we exclude the graph searching module in this phase of our research, as it can be influenced by the intermediate results of previous training. The results are presented in Table IV.

Our experimental results reveal that self-supervised models trained with either the instance discrimination task or the clustering task demonstrate comparable performance. However, when fine-tuned with only 3% of the data, these models exhibit significant variations in their results. This indicates that the instance discrimination task is more effective in feature extraction and plays a crucial role in training the feature extractor component. In contrast, the clustering task excels at emphasizing high-level information and promoting cluster stabilization.

TABLE IV
ABLATION STUDIES—PRETEXT TASKS

Backbone	Setting	SSL Task	OA	AA	AP	F1	kappa
ResNet-18	SSL	instance	0.48	0.47	0.51	0.47	0.40
		cluster	0.49	0.49	0.50	0.48	0.42
		both	0.52	0.57	0.53	0.52	0.44
	FT(3%)	instance	0.82	0.74	0.73	0.73	0.78
		cluster	0.73	0.66	0.63	0.65	0.68
		both	0.82	0.74	0.73	0.73	0.79
Inception_v3	SSL	instance	0.50	0.50	0.52	0.48	0.43
		cluster	0.50	0.49	0.56	0.49	0.43
		both	0.55	0.52	0.56	0.50	0.48
	FT(3%)	instance	0.91	0.82	0.80	0.81	0.89
		cluster	0.75	0.67	0.66	0.66	0.70
		both	0.91	0.90	0.93	0.91	0.90

TABLE V
ABLATION STUDIES—k IN KNN GRAPH SEARCHING

Method	k value	hit_avg	OA	AA	F1	Kappa	NMI
OA-SSL	base	-	0.54	0.58	0.53	0.46	0.44
	1	99%	0.56	0.63	0.55	0.50	0.51
	3	97%	0.56	0.63	0.55	0.50	0.52
	5	97%	0.56	0.63	0.54	0.49	0.51
	7	95%	0.56	0.63	0.55	0.50	0.52
	10	94%	0.56	0.63	0.55	0.50	0.52
GCC	base	-	0.37	0.39	0.37	0.28	0.25
	3	81%	0.38	0.40	0.38	0.29	0.27

3) Selection of k in KNN Graph Searching: We investigate the impact of different values of k in KNN on our proposed OA-SSL method. Our graph search mechanism relies on the assumption that the category of each of the k-KNNs is consistent with that of the original sample. We introduce the indicator hit_avg to assess the category consistency probability. We evaluate the GCC using single-source SAR and the OA-SSL using SAR with optical assistance (both based on ResNet18), and the results are presented in Table V. The results show that the effectiveness of graph search requires hit_avg to exceed a threshold of 90%. Due to the performance limitations of SAR image clustering, the hit_avg of GCC only reaches 80%. In contrast, in our OA-SSL, the hit_avg ranges from 95% to 99%, indicating more reliable positive samples obtained through graph searching, which leads to performance improvement.

Furthermore, our proposed method employs the matched intermediate domain image of the searched samples for the construction of positive sample pairs, rather than using SAR images directly. Since the KNN graph is constructed based on the features of SAR images, the searched samples obtained from graph searching within the same domain are highly similar to the original samples. Using these similar samples as positive pairs may not yield significant information gain and could potentially render instance-level pretext tasks ineffective. This could be another explanation for why the graph search

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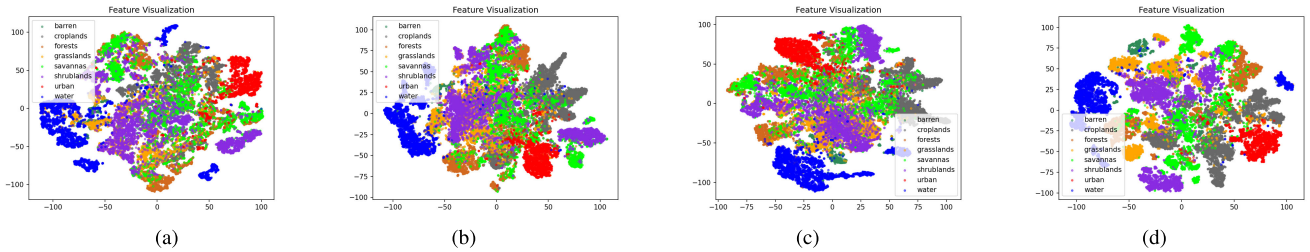


Fig. 9. Visualization of features using data from different domains as training auxiliary data. Using the generated image from optics to SAR as the intermediate domain enhances feature extraction, resulting in stronger class separability. (a) Only SAR (OA: 37%). (b) SAR and optical-RGB (OA: 46%). (c) SAR and optical-gray (OA: 48%). (d) SAR and optical-gen (OA: 56%).

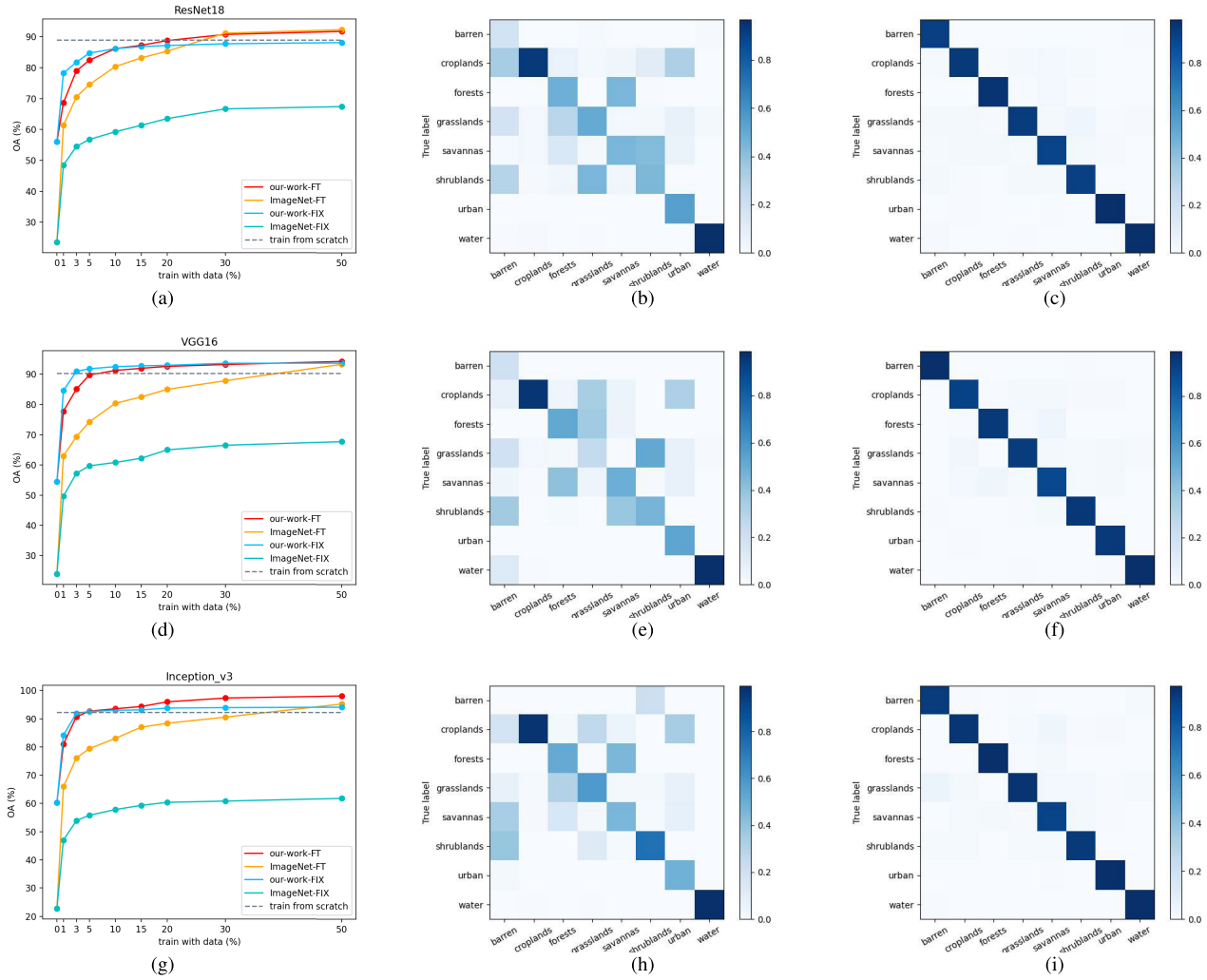


Fig. 10. Fine-tune results: the first column is the classification accuracy curve of the models fine-tuned with different scales of data, and our model demonstrates the ability to achieve significant performance improvements even with a small amount of data. The second column is the confusion matrix of the OA-SSL model before fine-tuning. Certain visually similar classes (such as shrublands and grassland) are difficult to separate by complete self-supervised clustering. The third column is the confusion matrix after fine-tuning where the vast majority of images can be correctly classified. (a) Fine-tune with $k\%$ data—ResNet. (b) SSL cluster—ResNet. (c) After fine-tuned—ResNet. (d) Fine-tune with $k\%$ data—VGG. (e) SSL cluster—VGG. (f) After fine-tuned—VGG. (g) Fine-tune with $k\%$ data—inception. (h) SSL cluster—inception. (i) After fine-tuned—inception.

module in our proposed method achieves greater performance improvements compared with the single-source GCC method.

The time of KNN graph construction will increase with the increase of k. We use the *faiss* library [45] to efficiently build KNN. The time for constructing KNN with $k = 3$ for 14 208 samples and 512-D features is about 2 s, and the average time for each epoch is less than 0.5 s, which is negligible. However, as k increases, the construction time of the KNN graph increases rapidly.

With $k = 10$, the construction time increases to 40 s, which does introduce a nonnegligible time cost. However, it is observed that a larger value of k does not necessarily result in a proportional increase in performance gain. Setting k too small

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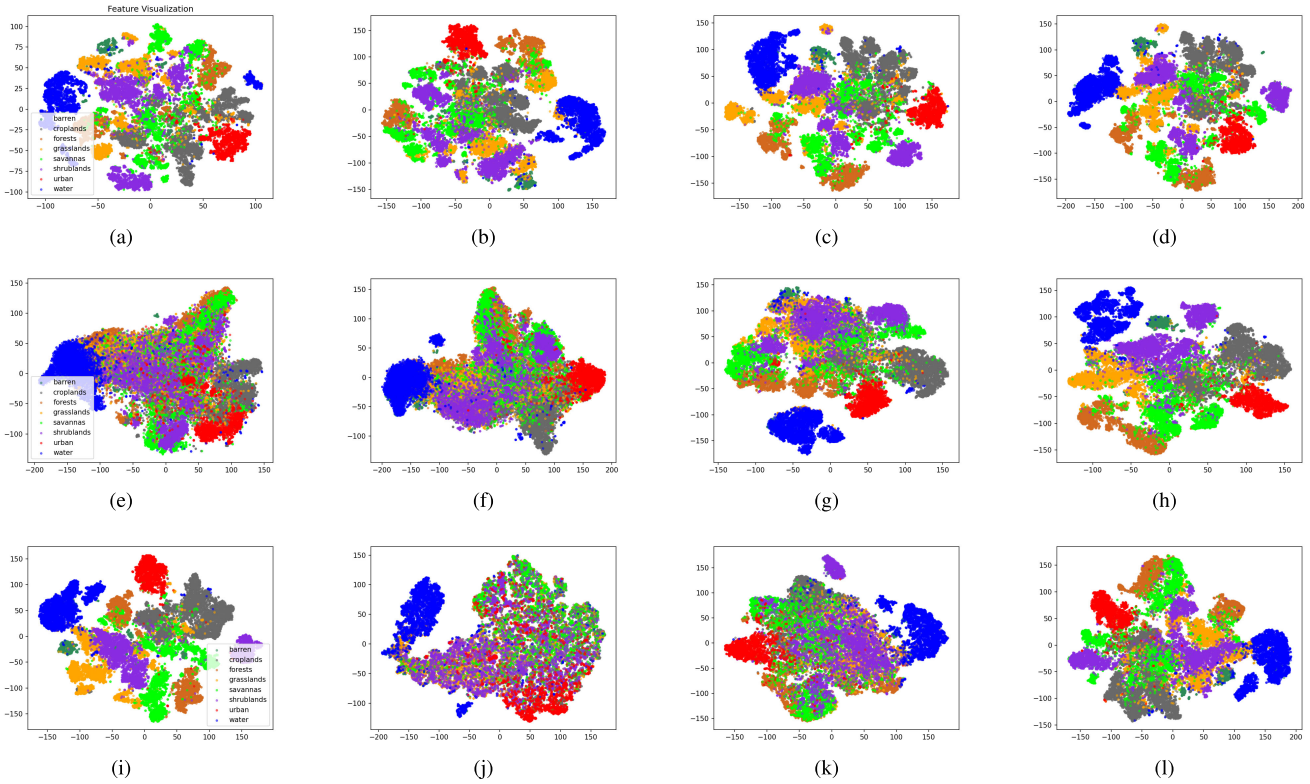


Fig. 11. Feature visualization. The first row: model pretrained with our method. The second row: model pretrained with ImageNet. The third row: model trained from scratch. The network architectures of the three are all Resnet18+MLP. Features extracted by our method are aggregated around some sample centers, and more easily to be fine-tuned. (a) OA-SSL (OA: 56.08%). (b) 1% data (OA: 68.63%). (c) 5% data (OA: 82.36%). (d) 15% data (OA: 87.17%). (e) ImageNet (OA: 23.62%). (f) 1% data (OA: 61.33%). (g) 5% data (OA: 74.51%). (h) 15% data (OA: 83.12%). (i) ResNet-18 70% data (OA: 92.8%). (j) 1% data (OA: 35.6%). (k) 5% data (OA: 56.6%). (l) 30% data (OA: 83.2%).

may lead to a fixed set of positive pairs, which can limit the expansion of positive samples. Therefore, to strike a balance between construction time and performance improvement, it is recommended to use a value of k, such as 3 or 5.

4) Selection of Cluster Number c: In the clustering task, we keep the clustering categories the same as the desired categories, enabling us to directly utilize the clustering results for obtaining classification results. However, when our model is a pretrained model aiming to enhance the feature extraction capabilities, we can increase the number of clusters, which can contribute to enhance feature extraction. To investigate this, we conduct experiments using the VGG backbone network. After pretraining the model with different cluster numbers, we freeze the parameters and set up two fully connected layers as classifiers, train the classifier using only 1% of the data (256 graphs), and test the classification results on the whole dataset. The experimental results are presented in Table VI.

If the cluster number is too small, the semantic features that the model focuses on may be too coarse, leading to a decline in performance. However, our model is not sensitive to the choice of c, indicating that it is not a task-specific parameter. The model's performance remains stable across a wide range of cluster numbers. As a pretrained feature extractor, it is recommended to set the cluster number c to be $3\times$–$5\times$ the actual number of categories.

TABLE VI
ABLATION STUDIES—CLUSTER NUMBER c

clusters	OA	AA	AP	F1	Kappa
3	0.8432	0.7550	0.7484	0.7503	0.8147
8	0.8459	0.8413	0.8779	0.8491	0.8186
15	0.8660	0.8661	0.8794	0.8710	0.8421
20	0.8737	0.8766	0.8816	0.8780	0.8516
30	0.8775	0.8793	0.8860	0.8815	0.8558
50	0.8772	0.8820	0.8909	0.8853	0.8556

F. Results After Fine-Tuned

Compared with the model trained from scratch or fine-tuning from the pretrained model on ImageNet dataset, our model can be fine-tuned with significantly fewer data ($\leq$15%) to achieve competitive results. Detailed results and confusion matrices are shown in Fig. 10.

The comparison of the confusion matrix before and after fine-tuning shows that the classification results of different network structures are similar, and they can all classify categories with sufficiently distinguishable features, such as croplands, urban, and water better. However, some categories with very similar features (e.g., grasslands, savannas, and shrublands) remain challenging to distinguish for self-supervised trained models. But supervised fine-tuning with a small number of samples can correct these errors easily.

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Due to optical image assistance, our model yields more refined and stable features. By keeping the number of clusters unchanged and using the ResNet18 backbone, we observed that fine-tuning the model with 20% of the data yields competitive classification results compared with training from scratch with 50% of the data. Similarly, with more complex models like Inception_v3, fixing the feature extractors and using only 3% of the data for fine-tuning is sufficient to achieve a satisfactory performance with a classification accuracy of 90%.

When comparing the SSL and fine-tuned results of different models, namely, ResNet18 (with fewer parameters), VGG16, and Inception_v3 (more complex models), we got several observations. First, a network with a smaller number of parameters tends to achieve better results through self-supervised training, but it may have limited and less stable feature extraction capabilities. On the other hand, more complex networks exhibit stronger feature extraction abilities, and our proposed model excels as a pretrained model. In experiments using VGG16 and Inception_v3 as backbone networks, our model consistently outperforms the models pretrained on ImageNet during fine-tuning, even surpassing the performance of the DF approach.

When comparing our model to ResNet18 pretrained with ImageNet and trained from scratch, we observed that our model achieves higher performance with fewer data for fine-tuning. This can be observed in the feature visualizations shown in Fig. 11. We hypothesize that this improved performance can be attributed to the following two factors.

1) Our optical-assistant SSL approach can obtain a better pretrained model, compared with models pretrained on natural image datasets like ImageNet. In addition, due to the self-supervised clustering task, the class separability of features extracted by our method is much stronger. As shown in Fig. 11(a), (f), and (k), although the classification accuracy of all three models is approximately 60%, our method produces significantly clearer features with fewer instances of different classes being confused with one another.
2) As seen in Fig. 11(a) and (e), due to instance-level contrastive loss and graph search, OA-SSL can extract features that are aggregated around some specific sample centers and exhibit a more compact distribution in clusters. This characteristic facilitates the fine-tuning of classifiers.

V. Conclusion

Considering the challenges of annotating large-scale SAR datasets, we propose a graph CC architecture that leverages geographically matched optical-SAR images. Our approach incorporates three types of positive pairs and employs instance-level contrastive and cluster-level contrastive SSL heads to extract informative features and facilitate land cover classification. We perform experiments on the SEN12MS dataset and demonstrate our OA-SSL model with optical image assistance learns more clustering friendly representations and outperforms other basic SSL methods. Additionally, the OA-SSL model can be fine-tuned with a small amount of data to achieve excellent classification performance, highlighting its practicality and scalability.

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